

Dental Infection Control

1. Key Definitions

Before diving into protocols, it is vital to understand the basic terminology:

- **Cross-Infection:** The transmission of infectious agents between patients and staff within a clinical environment.
- **Asepsis(Sterility):** The absence of pathogenic microorganisms.
- **Disinfection:** A process that eliminates many or all pathogenic microorganisms, except bacterial spores, on inanimate objects.
- **Sterilization:** The complete destruction of all forms of microbial life, including highly resistant bacterial spores.
- **Standard Precautions:** A set of infection control practices used to prevent transmission of diseases that can be acquired by contact with blood, body fluids, non-intact skin, and mucous membranes.

2. Common Infectious Diseases in Dentistry

Dental professionals operate in a high-risk environment due to exposure to blood and saliva. The most concerning pathogens include:

- **Hepatitis B (HBV) & Hepatitis C (HCV):** Highly resilient viruses transmitted through blood; HBV is preventable via vaccination.
- **Human Immunodeficiency Virus (HIV):** Transmitted through blood and certain body fluids.
- **Mycobacterium tuberculosis:** Transmitted via airborne droplets/aerosols.
- **Respiratory Viruses:** Including Influenza and SARS-CoV-2 (COVID-19).
- **Herpes Simplex Virus (HSV):** Transmitted through direct contact with lesions.

3. Routes of Transmission

Microorganisms in the dental office are transmitted through four primary routes:

1. **Direct Contact:** Touching blood, saliva, or lesions.
2. **Indirect Contact:** Touching contaminated instruments or surfaces.
3. **Droplet Spread:** Large particles (splatter) from coughing, sneezing, or talking that land on mucosal surfaces.
4. **Inhalation:** Fine aerosols generated by high-speed handpieces or ultrasonic scalers that remain suspended in the air.

4. Personal Protective Equipment (PPE)

PPE acts as a physical barrier to protect both the operator and the patient.

- ❖ **PPE Sequence:** Gown → Mask → Eyewear → Hand Hygiene → Gloves.
- ❖ **Hand Hygiene:** * **Routine:** 15–20 seconds with antimicrobial soap.
 - **Surgical:** 2–6 minutes scrubbing up to the elbows.
 - **Crucial Moments:** Before touching a patient, before aseptic procedures, and after body fluid exposure.
- **Gloves:** Must be changed between patients. Use heavy-duty utility gloves for cleaning instruments.
- **Masks:** High-filtration surgical masks (Level 2 or 3) or N95 respirators for aerosol-generating procedures.
- **Gowns:** Fluid-resistant protective clothing.
- **Protective Eyewear/Face Shields:** To protect against splatter and projectiles.

5. Classification of Dental Instruments (Spaulding's Classification)

Instruments are categorized based on the risk of infection they pose:

Category	Definition	Examples	Processing
Critical	Penetrates soft tissue or bone.	Scalpels, burs, forceps, scalers.	Sterilization
Semi-Critical	Touches mucous membranes but does not penetrate.	Mouth mirrors, impression trays.	Sterilization (High-level disinfection if heat-sensitive)
Non-Critical	Touches only intact skin.	X-ray cone, blood pressure cuff.	Low to intermediate disinfection

6. Sterilization and Disinfection Methods

A. Chemical Disinfection

Used for surfaces or heat-sensitive items.

- **High-level:** Glutaraldehyde (Cidex), Peracetic acid.
- **Intermediate:** Sodium Hypochlorite (Bleach), Iodophors.

B. Mechanical (Autoclave)

The **Steam Under Pressure (Autoclave)** is the gold standard for dental sterilization.

□ Standard **Cycle:** 121°C at 15 psi for 15–30 minutes.

□ Fast **Cycle:** 134°C at 30 psi for 3–10 minutes.

7. Monitoring and Calibration of Autoclaves

To ensure an autoclave is functioning correctly, three types of monitoring are required:

1. **Physical Monitoring:** Checking the pressure gauges and temperature displays during each cycle.
2. **Chemical Monitoring:** Internal and external indicators (autoclave tape) that change color when exposed to heat.
3. **Biological Monitoring (Spore Testing):** The most reliable method. Uses highly resistant spores (e.g., *Geobacillus stearothermophilus*). If the spores are killed, the sterilization process is successful. This should be performed at least weekly.

8. Hand Hygiene Technique

Handwashing is the single most important measure to reduce disease spread.

1. **Wet** hands with water.
2. **Apply** antimicrobial soap.
3. **Rub** hands together for at least 15–20 seconds (covering palms, backs of hands, between fingers, and under nails).
4. **Rinse** thoroughly.
5. **Dry** with a disposable paper towel and use the towel to turn off the faucet.

9. Chemical Dilution Protocols

Correct dilution is critical for efficacy and safety.

A. Sodium Hypochlorite (Bleach)

For general surface disinfection (usually 1:10 or 1:100 dilution):

- To get a **0.5% solution** from a **5% household bleach** concentrate: Mix **1 part bleach** with **9 parts water**.

B. Glutaraldehyde (Cidex/Sideq)

- Usually supplied as a 2% solution. It requires an **activator** (provided with the bottle). Once activated, it usually remains effective for 14 to 28 days.

10. Medical Waste Management

Waste must be segregated at the point of origin:

- **Sharps (Needles, scalpels, burs):** Must be placed in puncture-resistant, leak-proof **Sharps Containers**.
- **Infectious Waste (Blood-soaked gauze, tissue):** Must be placed in **Biohazard Bags** (usually red or yellow).
- **General Waste:** Non-contaminated paper, plastic, or office waste goes into regular trash.
- **Amalgam Waste:** Must be collected in special containers to prevent mercury environmental contamination.

11. Disinfection of the Dental Unit and Waterlines

The dental chair and its attached tubing (waterlines) are significant reservoirs for Biofilms.

A. Surface Disinfection (The Chair)

- **The "Spray-Wipe-Spray" Technique:** Spray the surface with an intermediate-level disinfectant, wipe it to remove organic debris, then spray again and leave it wet for the manufacturer's recommended contact time (usually 3–10 minutes).
- **Barriers:** Use disposable plastic covers on "high-touch" areas like light handles, control panels, and headrests.

B. Dental Unit Waterlines (DUWL)

- **Purging/Flushing:** Discharge water and air for **2 minutes** at the start of the day and for **20–30 seconds** between each patient to reduce the microbial load in the tubes.
- **Shock Treatment:** Use periodic chemical "shocks" (using tablets or specialized solutions like silver ions or hydrogen peroxide) to strip the biofilm from the internal pipes.
- **Anti-Retraction Valves:** Ensure handpieces have functional valves to prevent patient fluids from being sucked back into the unit lines.

12. Sodium Hypochlorite (Bleach): Dilution and Stability

Sodium Hypochlorite is unstable and degrades quickly once diluted.

- **Dilution Ratio:** For general surfaces, use a **1:100** dilution. For blood spills, use a **1:10** dilution.
- **Stability/Expiry:**
 - **Concentrated (Stock):** Usually stable for 6 months if stored in a cool, dark place.
 - **Diluted Solution:** It loses its potency rapidly. **Diluted bleach must be prepared fresh every 24 hours.** Any leftover solution from the previous day should be discarded.

13. Needle-Stick Injury (Post-Exposure Protocol)

If a dentist or staff member is punctured by a needle used on a patient, immediate action is required:

1. **Immediate First Aid:** * Wash the wound with soap and water immediately.
 - Do **not** scrub or squeeze the wound excessively; allow it to bleed freely under running water.
 - If blood splashes in the eyes/mouth, flush with large amounts of water or saline.
2. **Report & Document:** Report the incident to the clinic supervisor. Record the date, time, and patient's medical history.
3. **Source Testing:** If the patient is present, they should be asked (with consent) to be tested for HIV, HBV, and HCV.
4. **Post-Exposure Prophylaxis (PEP):** Go to a hospital or occupational health center immediately. For HIV, PEP is most effective when started within **2 hours** of exposure.

14. Hepatitis B Virus (HBV) Vaccination

HBV is the most infectious blood-borne pathogen in the dental setting.

- **Vaccination Schedule:** A 3-dose series (Month 0, Month 1, and Month 6).
- **Antibody Testing:** Perform a **Titer Test (Anti-HBs)** 1–2 months after the third dose. A level of >10 mIU/mL indicates lifelong immunity.
- **Booster (Renewal):** Routine boosters are generally **not recommended** for healthy individuals if the initial response was successful, as the immune system retains "memory."
- **Incubation Period:** 45 to 160 days (Average: 90 days).

15. HIV and HCV: Incubation and Blood Management

A. Incubation Periods

- **HIV:** Initial "flu-like" symptoms may appear in 2–4 weeks, but the virus can remain asymptomatic for years.
- **Hepatitis C (HCV):** 14 to 180 days (Average: 45 days). Note: There is currently **no vaccine** for HCV.

B. Management of Blood Spills

If blood is spilled on the floor or surfaces:

1. **Isolate:** Mark the area so no one walks through it.
2. **PPE:** Wear heavy-duty gloves, a mask, and a gown.
3. **Absorb:** Cover the spill with paper towels to soak up the liquid.
4. **Disinfect:** Pour **1:10 Sodium Hypochlorite** over the paper towels and let it sit for at least 10 minutes.
5. **Dispose:** Place all contaminated materials into a **Biohazard bag**.
6. **Final Wipe:** Clean the area again with fresh disinfectant.

16. Waste Management and Disposal

- **Sharps (Yellow Box):** Needles, burs, and blades. Never overfill beyond 3/4.
- **Biohazard (Red/Yellow Bags):** Blood-soaked gauze and extracted teeth.
- **General Waste (Black/Transparent Bags):** Office paper and plastic covers.